

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim: Simulate cross correlation and autocorrelation on discrete time signal	
Experiment No: 3	Date:	Enrollment No: 92510133028

EXPERIMENT -3

```

import numpy as np
import matplotlib.pyplot as plt

def cross_correlation(signal1, signal2):
    # Compute the cross-correlation
    cross_corr = np.correlate(signal1, signal2, mode='full')
    return cross_corr

def autocorrelation(signal):
    # Compute the autocorrelation
    auto_corr = np.correlate(signal, signal, mode='full')
    return auto_corr

# Define the discrete-time signals
signal1 = np.array([1, 2, 3, 4, 5])
signal2 = np.array([2, 4, 6, 8, 10])

# Compute the cross-correlation
cross_corr = cross_correlation(signal1, signal2)

# Compute the autocorrelation

auto_corr = autocorrelation(signal1)

# Plot the cross-correlation and autocorrelation signals
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.stem(cross_corr)
plt.title('Cross-correlation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')

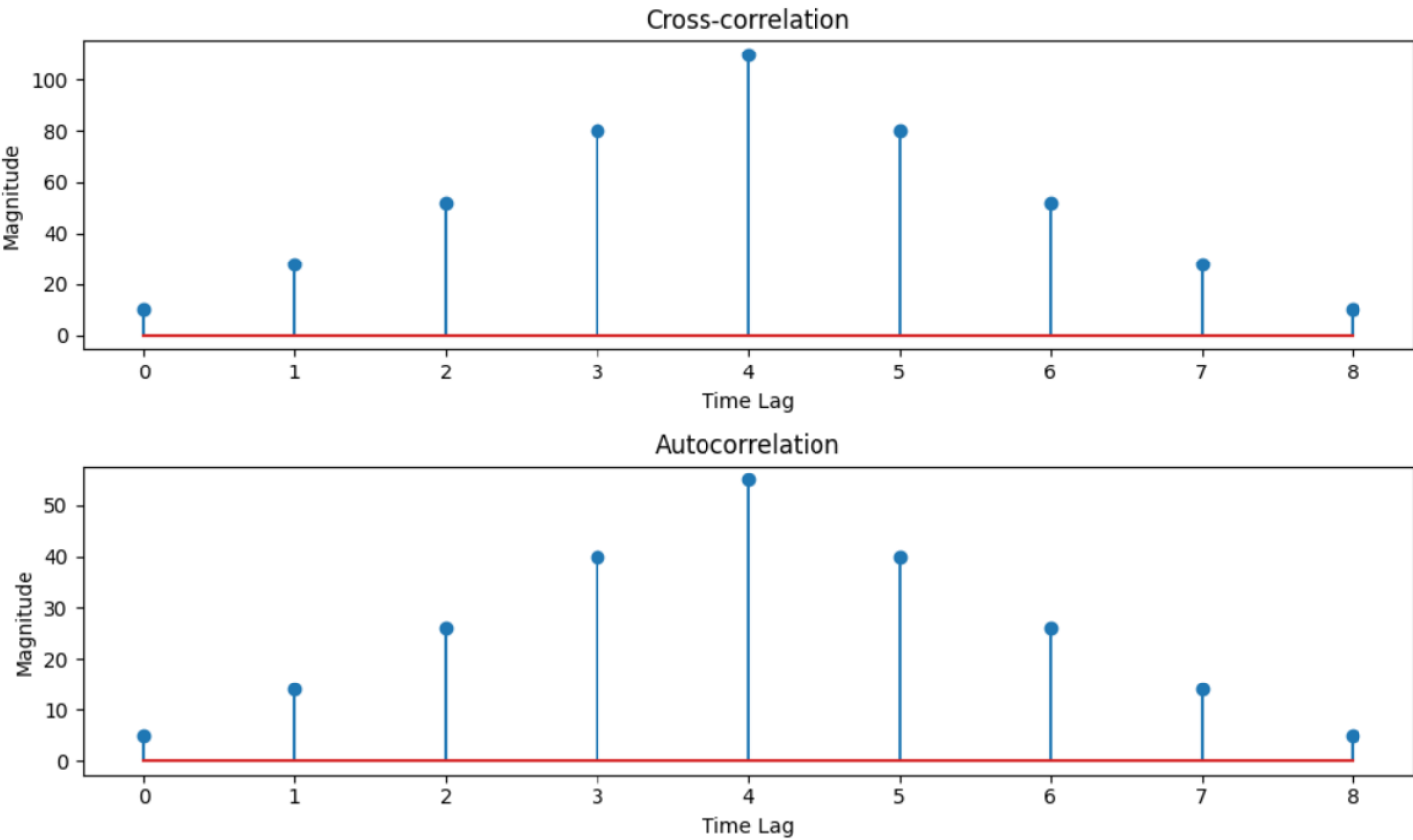
plt.subplot(2, 1, 2)
plt.stem(auto_corr)
plt.title('Autocorrelation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')

```

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```
plt.tight_layout()
plt.show()

# Save the cross-correlation or autocorrelation signals (optional)
# np.savetxt('cross_correlation.txt', cross_corr, delimiter=',')
# np.savetxt('autocorrelation.txt', auto_corr, delimiter=',')
```



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Conclusion:

The experiment was performed successfully. We calculated and plotted the cross-correlation and autocorrelation of discrete-time signals using Python. The graphs showed the similarity between two different signals (cross-correlation) and the similarity of a signal with itself (autocorrelation). This helped in understanding how correlation is used to compare signals in digital signal processing.

https://github.com/keshvi1234/DSIP_Exp